



NASA plans!

NASA announced it's plans to send in the SLS or Exploration Mission-1 with a dozen mini-satellites and Orion spacecraft in 2018. <http://dgit.in/SLSLaunch>

5G Drones

Drones won't be a menace anymore, can be proved by this new innovative Google project, 'Project Skybender' for 5G internet distribution. <http://dgit.in/5GDrone>

Space Age

OneWeb is of Great Britain while SpaceX is American, both the companies have the same goal.

It's not just achieving the objective that the firms have to focus on, they'll also have to contend with regulations regarding gaining of satellite spectrum. And since issues such as assigning satellite spectrum on such massive scales have never been tackled before by anyone on Earth, it's without the benefit of hindsight that authorities will have to make the necessary decisions.

All this makes it extremely interesting to keep a close watch on the way this new space race is unfolding.

And from what we could glean so far, it certainly seems as though Greg Wyler's OneWeb is on the right track. The company has not only earned the support of satellite makers such as Airbus, but also that of Virgin Galactic (Sir Richard Branson's blue eyed vision for the future) and also many other leading corporates and telecommunication providers. The fact that OneWeb is a startup makes this even more impressive. The company which is based in the Channel Islands of Britain intends to launch 600 to 700 satellites as soon as 2019.

The rosy picture of OneWeb doesn't mean Elon Musk's SpaceX is withdrawing from the race. SpaceX comes with its own expertise – the California-based aerospace company is not just one of the best rocket manufacturers around, but is also the fastest growing launch service provider in the world.

The company aims to launch about 4,000 satellites within the next five years. And with the backing of heavyweights such as Google and Fidelity who have already announced a \$1 billion investment for SpaceX, the firm has every reason to be hopeful of winning the latest space race.

Kuband and the supposed monopoly of OneWeb

One of the most fascinating things about satellite-based internet is that for it to operate, it will rely a lot on the Kuband radio frequency range. OneWeb was the first company to register its plans with the Geneva-based International Telecommunication Union (ITU) – a United Nations

agency for the coordination of satellite spectrum frequencies and orbital slots. This is a fact that OneWeb hastily let the world know. Richard Branson even suggested that this gave Onweb the claim over the entire Kuband frequency. To use his own words, OneWeb "has the rights, and there isn't space for another network."

But as maybe you've guessed from the fact that it's not the only active partici-

faithful. The consensus which the companies involved in the space race must come to is that they will have to draw from a reserve that none of them may own.

The actual bureaucratic hurdle comes with the fact that just like a country has its sovereign airspace, it also has spectrum control in its territory. This brings national-level bureaucracy into play, making it necessary for companies such as SpaceX



OneWeb's global web

pant of the space race, Branson's statements are hardly accurate. The truth is that although OneWeb may be the first to register interest, this does not grant it rights to a monopoly. It's highly unlikely that the United Nations is going to allow a monopoly, and besides, it's just better from a customer's standpoint to have competition – not just for prices, but also for quality of service.

When bureaucrats crash the party

As beautiful as the dream of bringing the internet to practically everyone on the planet is (and thereby making the world truly interconnected), the idea of satellite-based internet will have to contend with being handled by the bureaucracy before it can be realised. The ITU is just the beginning of the story. In fact, one needn't even consider the ITU as part of the bureaucracy per se. Contrary to what you think, the ITU isn't a patent office for spectrum. Instead, it's only a facilitator that ensures that the use of spectrum is equitable and

to gain the approval of distinct countries for the use of spectrum. Given the history of how even the most progressive of governments have elements in them that are corrupt, this could mean a major challenge for companies in the space race in the first half of the 21st century.

Technically speaking though, there are some technical challenges

Skeptics who view the satellite internet as a pipe dream of billionaires cite some technical problems (which are in turn linked to finance) as the reasons for their skepticism. And one of the main technical hassles they point out is called extreme latency. It's nothing but the difference in time between a satellite receiving a request and responding to it. This lag would obviously make such things as teleconferencing and online games impractical.

Musk and Wyler are both trying to get around this by deploying their satellites in what's termed as low Earth orbit – some 100 to 1,250 miles above Earth. By plac-